Staking and Validator Economics for Pars Network

Pars Network Technical Whitepaper PNP-013

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Abstract

We present the staking and validator economics for the Pars Network, extending the VEASHA tokenomics framework with concrete staking mechanisms, validator incentives, slashing conditions, delegation, and privacy-preserving stake proofs. Pars validators secure the privacy-preserving consensus layer (PNP-012) while maintaining anonymity—a requirement that conflicts with traditional staking designs where validator stakes are publicly visible. We introduce zero-knowledge stake proofs that demonstrate sufficient stake for validator eligibility without revealing the exact amount or the staker's identity. The economic model incentivises geographic distribution across diaspora jurisdictions, penalises correlation (validators in the same jurisdiction or hosting provider), and provides a delegation mechanism that preserves delegator privacy. Slashing conditions are designed for the coercion-resistant threat model: a validator under duress can trigger panic mode without being slashed for the resulting protocol deviation.

Keywords: staking, validator economics, VEASHA, slashing, delegation, zero-knowledge stake proofs, geographic distribution, coercion-resistant slashing

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1 Introduction

Proof-of-stake networks secure consensus through economic bonds: validators lock tokens as collateral, earning rewards for honest participation and risking slashing for protocol violations. The Pars Network's privacy requirements introduce unique constraints on staking design:

1. **Stake privacy.** Public stake amounts create power maps that adversaries use to identify high-value targets for coercion. Validator stakes must be hidden.
2. **Identity privacy.** Validators participate anonymously (PNP-012). Staking must not break anonymity.
3. **Coercion tolerance.** Slashing rules must accommodate panic-mode behaviour (PNP-005) without penalising coerced validators.
4. **Geographic distribution.** The diaspora is distributed across 60+ countries; staking incentives must encourage this distribution rather than concentrating in low-cost hosting regions.

1.1 Relation to VEASHA Tokenomics

The VEASHA (Vote-Escrow ASHA) tokenomics paper defines the token distribution, emission schedule, and governance voting power. This paper extends VEASHA with:

- Concrete staking and unstaking mechanics.
- Validator reward distribution.
- Slashing conditions and dispute resolution.
- Delegation with privacy preservation.
- Zero-knowledge stake eligibility proofs.

1.2 Contributions

- Zero-knowledge stake proofs for private validator eligibility.
- Correlation-penalised rewards encouraging geographic diversity.
- Coercion-aware slashing with panic-mode exceptions.
- Private delegation via stealth staking addresses.
- Formal analysis of economic security under the Pars threat model.

2 VEASHA Staking

2.1 Token Mechanics

ASHA is the native token of the Pars Network. VEASHA (vote-escrow ASHA) is obtained by locking ASHA for a period T:

Definition 2.1 (VEASHA Balance)**.** *A user locking a ASHA for period T (in weeks) receives VEASHA balance:*

$$\mathsf{veASHA}(a, T) = a \cdot \frac{T}{T_{\max}}$$

where $T_{\max} = 208$ weeks (4 years). VEASHA balance decays linearly as the lock approaches expiry.

2.2 Validator Registration

To become a validator, a participant must:

1. Lock at least $S_{\min}$ ASHA as VEASHA, where $S_{\min}$ is the minimum stake (initially 10,000 ASHA).
2. Generate a validator key pair and register the public key on the Pars L2 staking contract.
3. Generate a ZK proof of stake eligibility (Section 3).
4. Begin participating in consensus (PNP-012).

The registration transaction links the VEASHA position to the validator key via a commitment, without revealing the stake amount.

3 Zero-Knowledge Stake Proofs

3.1 Motivation

Traditional proof-of-stake systems publish validator stakes on-chain, enabling stake-weighted leader election and reward distribution. This transparency is a liability for Pars: an adversary monitoring the staking contract can identify the largest stakers and target them for coercion.

3.2 Construction

We use a Groth16 proof (infrastructure from PNP-006) to prove stake eligibility without revealing the amount.

Definition 3.1 (Stake Eligibility Proof)**.** *Given a VEASHA commitment $C = \mathsf{Commit}(a, T, r)$ on-chain, the validator produces a ZK proof π of the statement:*

$$\exists\, a, T, r : C = \mathsf{Commit}(a, T, r) \;\wedge\; a \cdot T / T_{\max} \geq S_{\min}$$

The proof reveals nothing beyond the fact that the committed VEASHA balance exceeds $S_{\min}$.

3.3 Stake Ranges for Weighted Sampling

Consensus (PNP-012) requires stake-weighted validator sampling. Revealing exact weights breaks privacy. Instead, we define stake ranges:

Validators prove membership in a tier via a range proof, revealing only the tier, not the exact amount.

Theorem 3.1 (Tier Privacy)**.** *Under the ZK proof soundness and zero-knowledge properties, the adversary learns only the validator's tier, not their exact stake. Within a tier, all validators are indistinguishable.*

Table 1: Stake ranges for weighted consensus sampling.

Tier	VEASHA Range	Sampling Weight
Bronze	$[S_{\min}, 3S_{\min})$	1
Silver	$[3S_{\min}, 10S_{\min})$	3
Gold	$[10S_{\min}, 30S_{\min})$	8
Diamond	$\geq 30S_{\min}$	15

4 Validator Rewards

4.1 Emission Schedule

Block rewards follow the VEASHA emission schedule: a decreasing emission rate starting at 5% annual inflation, halving every 4 years, with a hard cap of 1 billion ASHA.

Definition 4.1 (Block Reward)**.** *The reward for block b at epoch e is:*

$$R(b, e) = R_0 \cdot 2^{-e/E_{\text{half}}} \cdot \frac{1}{B_e}$$

where R_0 is the initial annual emission, E_{half} is the halving period in epochs, and B_e is the number of blocks in epoch e.

4.2 Reward Distribution

Rewards are distributed proportionally to sampling weight, adjusted by two factors:

Uptime factor.

$$f_{\text{up}}(V_i) = \frac{\text{blocks participated}}{\text{blocks eligible}} \cdot \mathbf{1}[f_{\text{up}} \geq 0.9]$$

Validators below 90% uptime receive zero rewards for that epoch.

Diversity factor.

$$f_{\text{div}}(V_i) = 1 - \alpha \cdot \frac{n_{\text{same}}(V_i)}{n_{\text{total}}}$$

where $n_{\text{same}}(V_i)$ is the number of validators sharing the same jurisdiction or AS number as V_i, and α is a correlation penalty parameter (initially $\alpha = 0.5$).

Definition 4.2 (Validator Reward)**.** *Validator V_i with tier weight w_i, uptime factor $f_{\text{up}}(V_i)$, and diversity factor $f_{\text{div}}(V_i)$ receives epoch reward:*

$$r_i = R_{\text{epoch}} \cdot \frac{w_i \cdot f_{\text{up}}(V_i) \cdot f_{\text{div}}(V_i)}{\sum_j w_j \cdot f_{\text{up}}(V_j) \cdot f_{\text{div}}(V_j)}$$

4.3 Geographic Distribution Incentive

The diversity factor penalises validator concentration. If 30% of validators are in a single jurisdiction, each receives a 15% reward reduction ($\alpha = 0.5$, $n_{\text{same}}/n_{\text{total}} = 0.3$). This incentivises validators to distribute across diaspora hubs.

Proposition 4.1 (Nash Equilibrium)**.** *Under the diversity-adjusted reward function, the Nash equilibrium validator distribution approximates the uniform distribution over available jurisdictions, weighted by the diaspora population distribution.*

5 Slashing

5.1 Slashable Offences

Table 2: Slashing conditions and penalties.

Offence	Evidence	Penalty
Double signing	Two signatures at same height	5%
Prolonged downtime (>48h)	Missing from >48h of blocks	1%
Equivocation	Conflicting votes in same round	10%
Censorship (proven)	Inclusion proof violation	3%
Correlated failure	>1/3 same-AS validators fail	2× base

5.2 Coercion-Aware Slashing

Standard slashing penalises protocol deviations indiscriminately. Under the Pars threat model, a coerced validator may deviate from the protocol involuntarily.

Definition 5.1 (Panic-Mode Exception)**.** *A validator in panic mode (PNP-005) whose protocol deviation matches the panic-mode behaviour profile is exempt from slashing. The panic mode produces:*

(i) Valid signatures under the decoy key (not equivocation).

(ii) Graceful withdrawal from consensus (not prolonged downtime if within 24h).

(iii) Alert to the guardian network (PNP-010) for key resharing.

Theorem 5.1 (Coercion Safety)**.** *A coerced validator who activates panic mode is not slashed, provided: (a) the panic credential is valid, (b) the deviation matches the panic behaviour profile, and (c) key resharing completes within 24h.*

5.3 Correlated Slashing

Correlated failures—multiple validators failing simultaneously due to shared infrastructure—indicate systemic risk. Pars applies a superlinear penalty:

$$\mathsf{slash}(V_i) = \mathsf{base_penalty} \cdot \min\left(3,\ 1 + \frac{|\{V_j : \text{correlated with } V_i\}|}{n/3}\right)$$

This incentivises infrastructure diversity: validators sharing a hosting provider or jurisdiction face amplified slashing risk.

6 Delegation

6.1 Privacy-Preserving Delegation

ASHA holders who do not wish to run validators can delegate their stake. Delegation must preserve delegator privacy: the delegator's identity, delegation amount, and choice of validator should not be publicly visible.

Definition 6.1 (Stealth Delegation)**.** *A delegator sends ASHA to a stealth staking address derived from the validator's public staking key, using the stealth address mechanism of PNP-007:*

(i) Delegator generates ephemeral key r and computes stealth address $A_s = H(r \cdot V_{\text{pub}}) \cdot G + V_{\text{stake}}$.

(ii) Delegator sends ASHA to A_s with a VEASHA lock.

(iii) Validator detects the delegation by scanning with their staking key.

(iv) The delegation is invisible to external observers.

6.2 Reward Distribution to Delegators

Validators distribute rewards to delegators via stealth payments:

1. The validator computes each delegator's share based on their delegation weight.
2. Rewards are sent to fresh stealth addresses derived from each delegator's viewing key.
3. The on-chain observer sees only transfers between unlinkable addresses.

6.3 Delegation Commission

Validators set a commission rate $c \in [0, 0.20]$ (max 20%). The commission is enforced by the staking contract:

$$r_{\text{delegator}} = r_{\text{total}} \cdot \frac{d_i}{\sum_j d_j + s_{\text{self}}} \cdot (1 - c)$$

where d_i is the delegator's stake and s_{self} is the validator's self-stake.

7 Unstaking and Withdrawal

7.1 Unbonding Period

Unstaking requires an unbonding period of 14 days. During unbonding:

- The stake remains subject to slashing for offences committed before the unstaking request.
- The validator is removed from the active set.
- No rewards accrue.

7.2 Emergency Unstaking

Under the coercion model, a validator under threat may need to unstake immediately. Emergency unstaking is permitted with:

- A 5% early withdrawal penalty (burned, not redistributed).
- A valid panic credential (PNP-005).
- Guardian network confirmation (PNP-010).

8 Economic Security Analysis

8.1 Cost of Attack

Definition 8.1 (Byzantine Threshold Cost)**.** *The cost for an adversary to control $f = \lfloor (n-1)/3 \rfloor$ validators is:*

$$C_{\text{attack}} = f \cdot S_{\min} \cdot P_{\text{ASHA}} + C_{\text{infra}}(f)$$

where P_{ASHA} is the market price of ASHA and C_{infra} is infrastructure cost.

With $n = 100$ validators and $S_{\min} = 10{,}000$ ASHA, the adversary must acquire and lock at least 330,000 ASHA, plus maintain 33 geographically distributed validator nodes.

8.2 Stake Concentration Bound

Theorem 8.1 (Diversity Equilibrium)**.** *Under the diversity-adjusted reward function with $\alpha = 0.5$, no rational validator operator controls more than $\lceil n/J \rceil$ validators, where J is the number of available jurisdictions.*

Proof. An operator adding a validator in a jurisdiction where they already operate k validators receives reward factor $1 - 0.5 \cdot (k+1)/n_{\text{local}}$. When k exceeds the jurisdiction's proportional share, the marginal reward becomes negative relative to operating in an unoccupied jurisdiction. □

8.3 Long-Range Attack Resistance

VEASHA's time-locked staking prevents long-range attacks: an adversary cannot unstake, sell tokens, and then attempt to rewrite history with worthless keys. The 14-day unbonding period exceeds the finality horizon.

9 Governance Integration

VEASHA balances determine governance voting power as defined in the VEASHA tokenomics paper. Staking extends this:

- **Validator governance.** Validators have additional governance rights: proposing protocol upgrades, adjusting staking parameters ($S_{\min}$, α, commission caps).
- **Delegator governance.** Delegators retain governance voting power proportional to their delegation, independent of the validator's vote.
- **Slashing governance.** Disputed slashing events are resolved by VEASHA-weighted governance vote via the DAO (PNP-001 governance).

10 Related Work

Ethereum PoS. Ethereum's proof-of-stake uses 32 ETH minimum stake with public validator identities. Pars differs in requiring validator anonymity and private stake amounts.

Penumbra. Penumbra [2] provides private staking on a Cosmos chain. Pars adopts a similar ZK stake proof approach but extends it with coercion-aware slashing and geographic diversity incentives.

Lux Staking. The Lux network's staking provides the validator registry infrastructure. Pars validators are registered as a subnet on Lux, inheriting the P-chain's validator management with Pars-specific privacy extensions.

Curve veTokenomics. Curve's vote-escrowed CRV model inspired VEASHA. Pars extends the model with ZK proofs over the lock position and staking-specific reward adjustments.

11 Conclusion

We have presented the staking and validator economics for the Pars Network, extending the VEASHA tokenomics with concrete mechanisms for private staking, validator rewards, coercion-aware slashing, and privacy-preserving delegation. Zero-knowledge stake proofs enable validator eligibility verification without revealing stake amounts, while geographic diversity incentives encourage validator distribution across the diaspora. The coercion-aware slashing framework accommodates the unique threat model of communities operating under authoritarian surveillance, ensuring that validators under duress are not economically punished for activating panic protocols.

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